

Problems in Normal Distribution

TEST YOUR KNOWLEDGE

1. The mean height of 500 male students in a certain college is 151 cm and the standard deviation is 15 cm. Assuming the heights are normally distributed, find how many students have heights between 120 and 155 cm ?
2. An aptitude test for selecting officers in a bank is conducted on 1000 candidates. The average score is 42 and the standard deviation of score is 24. Assuming normal distribution for the scores, find
 - (i) The number of candidates whose scores exceed 60
 - (ii) The number of candidates whose scores lie between 30 and 60.
3. In a normal distribution, 7% of the items are under 35 and 89% are under 63. What are the mean and standard deviation of the distribution ?
4. Let X denote the number of scores on a test. If X is normally distributed with mean 100 and standard deviation 15, find the probability that X does not exceed 130.
5. It is known from the past experience that the number of telephone calls made daily in a certain community between 3 p.m. and 4 p.m. have a mean of 352 and a standard deviation of 31. What percentage of the time will there be more than 400 telephone calls made in this community between 3 p.m. and 4 p.m. ?
6. Students of a class were given a mechanical aptitude test. Their marks were found to be normally distributed with mean 60 and standard deviation 5. What per cent of students scored
 - (i) more than 60 marks ? (ii) less than 56 marks ? (iii) between 45 and 65 marks ?
7. In an examination taken by 500 candidates, the average and the standard deviation of marks obtained (normally distributed) are 40% and 10%. Find approximately :
 - (i) How many will pass, if 50% is fixed as a minimum ?
 - (ii) What should be the minimum if 350 candidates are to pass ?
 - (iii) How many have scored marks above 60% ?

Answers

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| 1. 300 | 2. (i) 252 (ii) 533 | 3. $\bar{x} = 50.3, \sigma = 10.33$ |
| 4. 0.9772 | 5. 6.06% | 6. (i) 50% (ii) 21.2% (iii) 84% |
| 7. (i) 79 (ii) 35% (iii) 11. | | |